

ASSIGNMENT 3

Title/Objective : Embedded C program for memory Block Transfer

Problem Statement : Write an Embedded C program to transfer elements from one location to another for following :-

- i) Internal to internal memory transfer
- ii) Internal to external memory transfer

Input:

- i) Internal to internal memory transfer

```
#include<xc.h>

void main(void)
{
    int i;

    int arr1[]={0x21,0x22,0x23,0x24,0x25};
    int arr2[]={0x00,0x00,0x00,0x00,0x00};

    for(i=0 ; i<4 ; i++)
    {
        arr1[i]=arr2[i];
    }

    return;
}
```

- ii) Internal to external memory transfer

```
#include<xc.h>

void main(void)
{
    int i,temp[];

    int a[]={0x21,0x22,0x23,0x24,0x25};
    int b[]={0x99,0x99,0x99,0x99,0x99};
```

```
#include <xc.h>
```

```
void main(void)
{
    int i;
    int temp[5]; // Define size

    int a[] = {0x21, 0x22, 0x23, 0x24, 0x25};
    int b[] = {0x99, 0x99, 0x99, 0x99, 0x99};

    for(i = 0; i < 5; i++) // Loop through all 5 elements
    {
        temp[i] = a[i];
        a[i] = b[i];
        b[i] = temp[i];
    }

    while(1); // Optional: infinite loop to prevent program
    exit
}
```

Output:

File Registers x																		
	Address	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	ASCII
	000	00	00	00	00	00	00	00	00	00	00	00	21	00	22	00	23	!.".#
	010	00	24	00	25	00	01	00	00	00	00	00	00	00	00	00	00	.\$.%.
	020	00	21	00	22	00	23	00	24	00	25	00	00	00	00	00	00	!.".#.\$.%....
	030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
	040	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
	050	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
	060	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
	070	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
	080	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
	090	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
	0A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
	0B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
	0C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
	0D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	

Memory File Registers Format Hex

Filter Address

File Registers

Address	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	ASCII
000	00	21	00	22	00	23	00	24	00	00	00	21	00	22	00	23	.! ".#\$ %&'()*+,-./:;<=>?@ [^_`{ }~
010	00	24	00	25	00	21	00	99	00	99	00	99	00	99	00	01	.\$.%!..
020	00	99	00	99	00	99	00	99	00	99	00	21	00	22	00	23	 !".#
030	00	24	00	25	00	00	00	00	00	00	00	00	00	00	00	00	.\$.%.....
040	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
050	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
060	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
070	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
080	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
090	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
0D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	

Memory File Registers Format Hex

Filter Address